

HNSS03 · Pharyngeal Arch Birth Defects — Pathology Chart

Sorted by what failed: **cleft**, **pouch**, **neural crest**, or a **teratogen** hitting neural crest | **STEP 1** tags mark content beyond the lecture · **EXAM EMPHASIS**
= flagged in class as testable (branchial fistula slide, CATCH-22, alcohol-induced oxidative stress)

HoyaDocs
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Cleft & Pouch Defects						
DEFECT	WHO / CAUSE	WHAT FAILED (EMBRYOLOGY)	KEY FINDINGS	GENE / MOLECULAR	DIAGNOSIS & BUZZWORDS	MANAGEMENT & KEY FACTS
Branchial fistula / lateral cervical cyst <i>the cleft that didn't close</i> CLEFT 2 EXAM EMPHASIS	Congenital; children / young adults	Cleft 2 fails to grow caudally over clefts 3–5 → cervical sinus persists ; if a canal stays open to the skin → branchial fistula	Painless, fluctuant lateral neck mass along the anterior border of the SCM ; may drain or enlarge with infection	—	Does not move with swallowing. Histology: squamous lining + lymphoid tissue (HNSS16)	Excision STEP 1 . Contrast thyroglossal duct cyst: midline , moves with swallowing
DiGeorge syndrome (sequence) <i>the pouches that didn't form</i> POUCHES 3 & 4 EXAM EMPHASIS	Human: 22q11.2 deletion STEP 1 ; mouse model: Hoxa-3 knockout	Pouches 3 and 4 malform → no thymus (pouch 3 ventral) and no parathyroids (pouch 3 dorsal + 4 dorsal)	Absent thymus → T-cell deficiency (viral/fungal infections) · absent parathyroids → hypocalcemia (tetany, seizures) · abnormal ears, hypertelorism	22q11.2 (TBX1) STEP 1 ; Hoxa-3 –/– mice look very similar	CATCH-22 : Cardiac (truncus, tetralogy), Abnormal face, Thymic hypoplasia, Cleft palate, Hypocalcemia. No thymic shadow ; ↓ PTH	Calcium/vitamin D; cardiac repair; thymus transplant STEP 1

★ **USMLE high-yield**: Glands come from **pouches**; a question that pairs **hypocalcemia + infections + no thymic shadow** is pouches 3 & 4. A **lateral** neck cyst anterior to the SCM is **cleft 2**.

Neural Crest Syndromes — neural crest lacks SOD & catalase						
DEFECT	WHO / CAUSE	WHAT FAILED (EMBRYOLOGY)	KEY FINDINGS	GENE / MOLECULAR	DIAGNOSIS & BUZZWORDS	MANAGEMENT & KEY FACTS
Treacher Collins syndrome <i>the first arch that didn't fill in</i> ARCH 1 NC	Inherited (autosomal dominant STEP 1); very high retinoic acid in mice produces a similar phenotype	NC migration from R2/R4 into arches 1 (and 2) fails (~wk 2–3.5)	Malformed zygomatic bones, mandibular hypoplasia, lower eyelid colobomas, malformed external ears	Tcof-1 mutation	Cheek + jaw + ear bones = all arch 1; conductive hearing loss (ossicles) STEP 1 ; normal calcium, thymus, pigment	Airway risk (small jaw) and surgical reconstruction STEP 1
Waardenburg syndrome <i>the pigment-and-hearing crest defect</i> NC GENE	Inherited (autosomal dominant STEP 1)	Neural crest defect → melanocytes and inner-ear contributions affected	White forelock (white lock on dark hair), deafness, ocular hypertelorism, cleft palate	PAX3 mutation	"White forelock + Wide eyes + Won't hear"; heterochromia STEP 1	The only one of these with a pigment change
Neurofibromatosis <i>the neural crest tumors</i> NC TUMORS	Inherited (autosomal dominant) STEP 1	Tumors of neural crest origin	Neurofibromas ; café-au-lait spots, Lisch nodules (NF1) STEP 1	NF1 (chr 17), NF2 (chr 22) STEP 1	NF2 → bilateral vestibular schwannomas STEP 1	Lecture: know that it exists as a neural crest disorder

★ **USMLE high-yield**: Treacher Collins = **arch 1 neural crest** (Tcof-1). Waardenburg = **PAX3** (pigment + deafness). Don't confuse Treacher Collins' **lower-eyelid** coloboma with an **iris** coloboma (choroid fissure, HNSS18).

Teratogen Embryopathies — toxins that hit neural crest

DEFECT	WHO / CAUSE	WHAT FAILED (EMBRYOLOGY)	KEY FINDINGS	GENE / MOLECULAR	DIAGNOSIS & BUZZWORDS	MANAGEMENT & KEY FACTS
Fetal alcohol syndrome <i>the flattened midface</i> 1 HIT EXAM EMPHASIS	Maternal alcohol ; dose-dependent	Alcohol → oxidative stress (ROS) → damages/kills NC cells, which lack SOD and catalase	Smooth philtrum, thin upper lip, short palpebral fissures, microcephaly , intellectual disability, growth restriction STEP 1	↑ F2-isoprostanes (marker of oxidative stress): +69% / 289% / 345% at blood alcohol 0.08 / 0.10 / 0.13	" Alcohol hits once " (ROS only)	Leading preventable cause of intellectual disability STEP 1 ; vitamin C lowered F2-isoprostanes in the study
Isotretinoin / retinoic acid embryopathy <i>ears, jaw and heart</i> 2 HITS	High-dose retinoic acid (isotretinoin, vitamin A)	① ROS damage to NC + ② RA floods RAREs in Hox promoters → Hox hyperactivation → scrambled Hox code	Microtia/anotia , small jaw, cleft palate, conotruncal heart defects , CNS defects, thymic hypoplasia STEP 1	RARE → Hox genes (too much Hox protein)	" RA hits twice "; in mice → Treacher Collins–like pups	Two forms of contraception (iPLEDGE) STEP 1

Know that these exist

DEFECT	WHAT TO KNOW (LECTURE: "KNOW THAT THEY EXIST")
Hallermann-Streiff syndrome	Similar to Treacher Collins (first-arch craniofacial defect)
Goldenhar syndrome	Craniofacial defect of arches 1–2; oculo-auriculo-vertebral spectrum (hemifacial microsomia, ear tags, epibulbar dermoids) STEP 1
Craniofacial microsomia	Underdevelopment of one side of the face (arch 1–2 structures)
Mandibulofacial dysostosis	Small mandible + malar (cheek) hypoplasia — the descriptive name for the Treacher Collins phenotype STEP 1
Meckel syndrome	Not Meckel's cartilage or Meckel's diverticulum! (Meckel–Gruber: encephalocele, polycystic kidneys, polydactyly) STEP 1
Cleft lip / cleft palate	Covered in detail in HNSS04: cleft lip = maxillary + medial nasal; cleft palate = palatine shelves

★ **USMLE high-yield: Name the drug first, then match the picture:** smooth philtrum → alcohol; microtia + conotruncal defect → isotretinoin. Other teratogens: thalidomide (limbs), valproate (neural tube), phenytoin (fetal hydantoin), lithium (Ebstein).